{EPITECH}

2025

# PROJECT SPECIFICATIONS

# EPITRELLO

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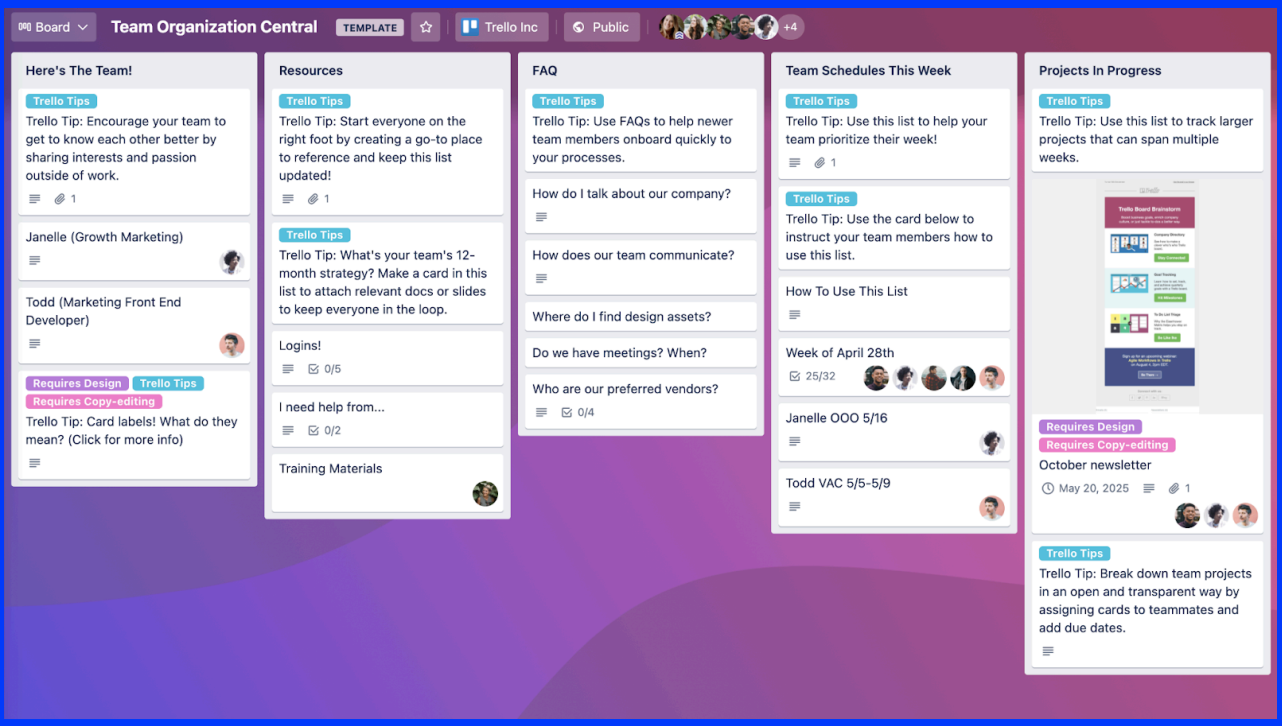
# 1. OVERVIEW

Before introducing the project itself, let’s put a little bit of context on this. For our third year of studies in computer science at Epitech, we must have practical work experience at the start of the year part-time (Thursdays and Fridays). This translates into a part-time internship or a professional project in a simulated work situation.

And this is where this project comes in. In the simulated professional work, we had to choose a project from a catalogue that the school provided. The projects refer to an existing application, and our goal is to stick as closely as possible to the functionalities of the original app.

About the app, I have chosen the EpiTrello project from the given projects list. Trello is an online project management tool, inspired by Toyota's Kanban method. It is based on the organization of projects into boards listing cards, each representing tasks. Cards are assignable to users and are movable from one board to another, representing their progress. Code name: “**Taskly**”.




Example of a Trello board

# 2. OBJECTIVES

- Create a visually intuitive and responsive Kanban board interface.
- Allow users to manage tasks through cards, lists, and boards.
- Support multiple users and shared boards.
- Allow real-time updates when multiple users interact with the same board.

# 3. CORE FEATURES

## 3.1. WEBSITE STRUCTURE

- Main section explaining the project (showcase)
- App '/app' for the application itself

## 3.2. USER MANAGEMENT

- User Registration & Login
  - Email + password authentication
  - Password reset functionality if forgotten
- User Profiles
  - Editable name, avatar, and bio

## 3.3. BOARDS

- Create / Edit / Delete boards
- Board sharing
  - Add/remove collaborators
  - Define permissions (Owner, Editor, Viewer)
- Board background customization
  - Custom Color or default image backgrounds
  - Import custom images (Optional)

## 3.4. LISTS

- Create / Rename / Delete lists
- Lists are ordered horizontally on the board
- Drag & Drop reordering of lists

## 3.5. CARDS (TASKS)

- Create / Edit / Delete cards
- Assign cards to users
- Add descriptions, due dates, and labels
- Comments section for discussions
- Drag & Drop cards between lists
- Card activity log

## 3.6. LABELS & FILTERS

- Custom color-coded labels per board
- Search/filter cards by:
  - Label
  - Assigned user
  - Keyword
  - Due date

## 3.7. REAL-TIME COLLABORATION

- Instant updates on:
  - Card movement
  - List creation
  - Comments
  - Member additions

### 3.8. NOTIFICATIONS

- In-app notifications for:
  - Updates to tasks you're assigned to
  - New comments
  - Due date reminders

## 4. TECHNICAL REQUIREMENTS

### 4.1. FRONTEND

- Framework: React
- Styling: Tailwind CSS
- HTTP Client: Axios

### 4.2. BACKEND

- Language: Python
- API Framework: FastAPI
- Database: MySQL with MariaDB server
- Authentication: JWT

### 4.3. DEPLOYMENT

- Containerization: Docker
- Hosting: Private debian server
- CI/CD: GitHub Actions

## 5. NON-FUNCTIONAL REQUIREMENTS

- Performance: Boards should load within 2 seconds
- Scalability: Support 100+ simultaneous users per board
- Security: Hash passwords (bcrypt), secure JWTs
- Responsiveness: Works on desktop, tablet, and mobile
- Docs:
    - README with installation & usage instructions
    - API documentation (FastAPI builtin docs)
    - In-code documentation

## 6. OPTIONAL ENHANCEMENTS

- Dark mode
- Calendar view for cards with due dates
- File attachments in cards
- Public board templates
- Offline mode

## 7. RESOURCES

- Original project website: https://trello.com
- TailwindCSS: https://tailwindcss.com/
- FastAPI: https://fastapi.tiangolo.com/
- Axios: https://axios-http.com/fr/